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UI/UX

Shneiderman’s 8 Golden Rules of UI Design

1. **Strive for Consistency**
2. **Enable Frequent Users to Use Shortcuts**
3. **Offer Informative Feedback**
4. **Design Dialogs to Yield Closure**
5. **Offer Simple Error Handling**
6. **Permit Easy Reversal of Actions**
7. **Let the User Be in Control**
8. **Reduce Short-Term Memory Load on the User**

1. Strive for Consistency

Answer:

UI components design must be of the **same style, theme, position and structure** so that it is easy for the users to use and remember.

Example:

Facebook uses a similar layout and navigation style on both desktop and mobile applications.

2. Enable Frequent Users to Use Shortcuts

Answer:

This rule means that experienced users should have shortcuts to perform tasks faster and reduce the number of interactions. Shortcuts, function keys, and hidden

commands improve efficiency for frequent users.

Example:

Microsoft Word provides shortcuts like **Ctrl + S** to save a document quickly.

3. Offer Informative Feedback

Answer:

Every user action should receive appropriate system feedback.

For frequent and small actions, simple feedback is enough, while important actions require detailed feedback.

Example:

When uploading a file, the system shows a progress bar to indicate the upload status.

4. Design Dialogs to Yield Closure

Answer:

A sequence of actions should have a clear beginning, middle, and end.

Users should receive feedback after completing a task so they understand that the operation is finished.

Example:

After completing an online payment, the system displays a "Payment Successful" confirmation message.

5. Prevent Errors

Answer:

The system should prevent serious errors and provide simple, understandable solutions when errors occur.

Error messages should clearly explain the problem and how to fix it.

Example:

A login page shows "Password must contain at least 8 characters" instead of only showing "Invalid input."

6. Permit Easy Reversal of Actions

Answer:

A good UI should allow users to undo or reverse their actions easily.

This reduces user anxiety and encourages users to explore the system freely.

Example:

Gmail provides an "Undo Send" option after sending an email.

7. Let the User Be in Control

Answer:

Users should feel that they are controlling the system rather than the system controlling them.

The system should respond to user actions and allow users to initiate tasks.

Example:

A video player allows users to **play, pause, stop, or skip** whenever they want.

8. Reduce Short-Term Memory Load

Answer:

The system should reduce the amount of information users need to remember while performing tasks.

The interface should keep information visible and avoid unnecessary complexity.

Example:

A website displays saved addresses during checkout instead of asking users to remember and type them again.

*Usability Heuristics

1. Visibility
2. Freedom
3. Error Prevention
4. Flexibility
5. Error Recovery
6. Mapping
7. Consistency
8. Recognition
9. Minimalism
10. Help

1. Visibility

Answer:

Visibility of system status means that the system should always inform users about what is happening through proper feedback.

It helps users understand the current state of the system and know whether their action is completed or still in progress.

Example:

When uploading a file, a progress bar shows the upload percentage and remaining time.

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2. Freedom (User Control and Freedom)

Answer:

Freedom means the system should provide good defaults and allow users to undo or reverse their actions.

It gives users control over their activities and helps them recover from mistakes.

Example:

In Gmail, users can use the "Undo Send" option after sending an email.

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3. Error Prevention

Answer:

Error prevention means the system should help users avoid making mistakes by providing proper guidance and restrictions.

A good interface prevents errors before they happen instead of only showing error messages afterward.

Example:

A registration form does not allow submission until all required fields are filled correctly.

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4. Flexibility

Answer:

Flexibility means the system should make advanced tasks more fluid and efficient while supporting different types of users.

It allows experienced users to perform tasks faster using shortcuts or advanced options.

Example:

Microsoft Word provides keyboard shortcuts like Ctrl + C and Ctrl + V for faster operations.

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5. Error Recovery

Answer:

Error Recovery means the system should help users recognize, diagnose, and recover from errors.

Error messages should be clear and provide guidance to solve the problem.

Example:

A login system displays "Incorrect password. Try again" instead of showing only "Login failed".

6. Mapping

Answer:

Mapping means the system should use familiar metaphors, symbols, and language that users can easily understand.

The interface should follow real-world concepts so that users can interact with the system naturally.

Example:

A shopping cart icon on an e-commerce website represents adding products before purchasing.

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7. Consistency

Answer:

Consistency means the system should use the same interface design, terminology, and behavior throughout the application.

It helps users learn the system easily and reduces confusion.

Example:

Facebook maintains similar navigation and button styles across different devices.

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8. Recognition

Answer:

Recognition means the system should make information easy to discover instead of forcing users to remember information from previous steps.

Example:

Google search suggestions show previous or related searches so users can select instead of typing everything again.

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9. Minimalism

Answer:

Minimalism means the interface should provide only necessary information in an elegant and simple way. Removing unnecessary elements helps users focus on important tasks.

Example:

A login page only contains necessary fields like Email and Password instead of unnecessary text and images.

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10. Help

Answer:

Provide help for users

Example:

A mobile application provides a step-by-step tutorial when a user opens a new feature for the first time.